

Please check the examination details below before entering your candidate information

Candidate surname

Other names

Pearson BTEC
Level 3
Nationals
Certificate

Centre Number

--	--	--	--	--	--

Learner Registration Number

--	--	--	--	--	--	--	--

Tuesday 15 January 2019

Afternoon (Time: 40 minutes)

Paper Reference **31617H/1B**

Applied Science / Forensic and Criminal Investigation

Unit 1: Principles and Applications of Science I
Biology

SECTION A: STRUCTURES AND FUNCTIONS OF CELLS AND TISSUES

You must have:

A calculator and a ruler.

Total Marks

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and learner registration number.
- Answer **all** questions.
- Answer the questions in the spaces provided
– *there may be more space than you need.*

Information

- The exam comprises three papers worth 30 marks each.
Section A: Structures and functions of cells and tissues (Biology).
Section B: Periodicity and properties of elements (Chemistry).
Section C: Waves in communication (Physics).
- The total mark for this exam is 90.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

P53979A

©2019 Pearson Education Ltd.

1/1/1/1/1/1/C2/



P 5 3 9 7 9 A 0 1 1 2

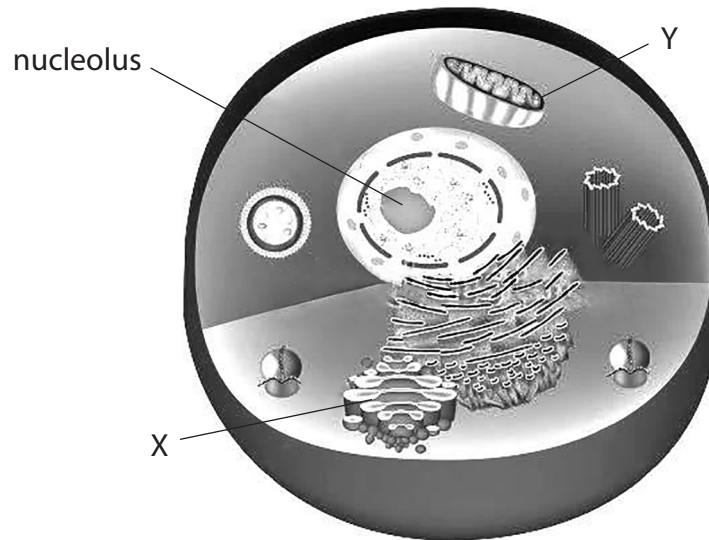


Pearson

Answer ALL questions. Write your answers in the spaces provided.

Some questions must be answered with a cross in a box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

- 1 Figure 1 shows the ultra-structure of an animal cell.



© Timonina/Shutterstock

Figure 1

- (a) (i) Name organelle X in Figure 1.

(1)

- (ii) Identify organelle Y in Figure 1.

(1)

- ☐ **A** centriole
- ☐ **B** endoplasmic reticulum
- ☐ **C** mitochondrion
- ☐ **D** vesicle

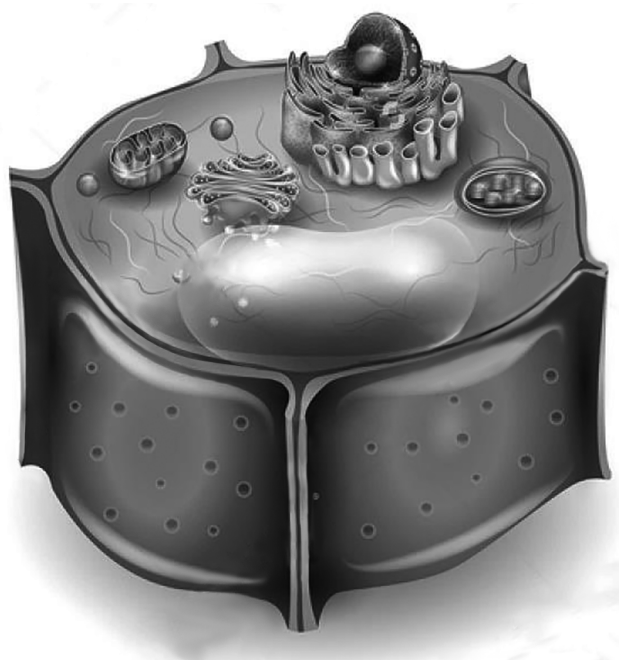


(iii) Identify the function of the nucleolus.

(1)

- ☐ A forms spindle fibres during cell division
- ☐ B makes RNA and ribosomes
- ☐ C regulates cellular activity
- ☐ D synthesises and transports lipids and carbohydrates

(b) Figure 2 shows the ultra-structure of a plant leaf cell.



© Tefi/Shutterstock

Figure 2

(i) Name **two** structural features that are found only in plant leaf cells and not in animal cells.

(2)

- 1
- 2

(ii) Name **one** structural feature that is found only in animal cells and not in plant leaf cells.

(1)

.....

(Total for Question 1 = 6 marks)



2 A white blood cell is a specialised cell.

(a) Complete the meaning of the term **specialised cell**.

(2)

A cell becomes specialised when its structure is altered. This enables a cell to have a function. This process is called cellular

(b) A neutrophil is a type of white blood cell.

Figure 3 shows neutrophils in a capillary.

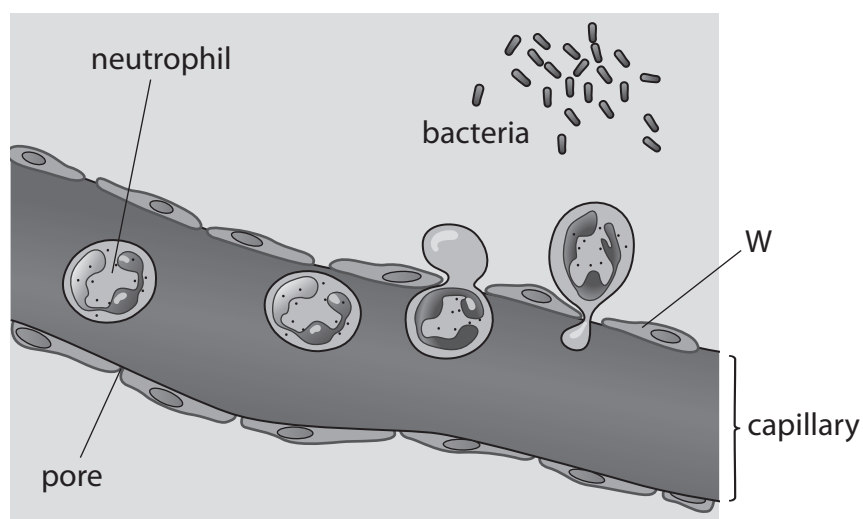


Figure 3

(i) Identify the type of cell labelled W in Figure 3.

(1)

- ☐ A ciliated
- ☐ B columnar
- ☐ C endothelial
- ☐ D stratified



Neutrophils travel in the blood. They move through pores in capillary walls to reach the bacteria in surrounding tissue.

- (ii) Explain how the structure of a neutrophil enables it to exit the capillary in order to reach the bacteria.

(2)

.....

.....

.....

.....

- (iii) Neutrophils contain many lysosomes.

Explain the function of lysosomes in a neutrophil.

(3)

.....

.....

.....

.....

.....

.....



P 5 3 9 7 9 A 0 5 1 2

- (c) During a bacterial infection, a person has a higher neutrophil count than before they were infected.

	neutrophil count per dm^3
before infection	3.5×10^9
during infection	6.1×10^{10}

Table 1

Calculate, using information in Table 1, the percentage increase in the neutrophil count during the infection.

Show your working.

(3)

percentage increase in the neutrophil count = %

(Total for Question 2 = 11 marks)



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

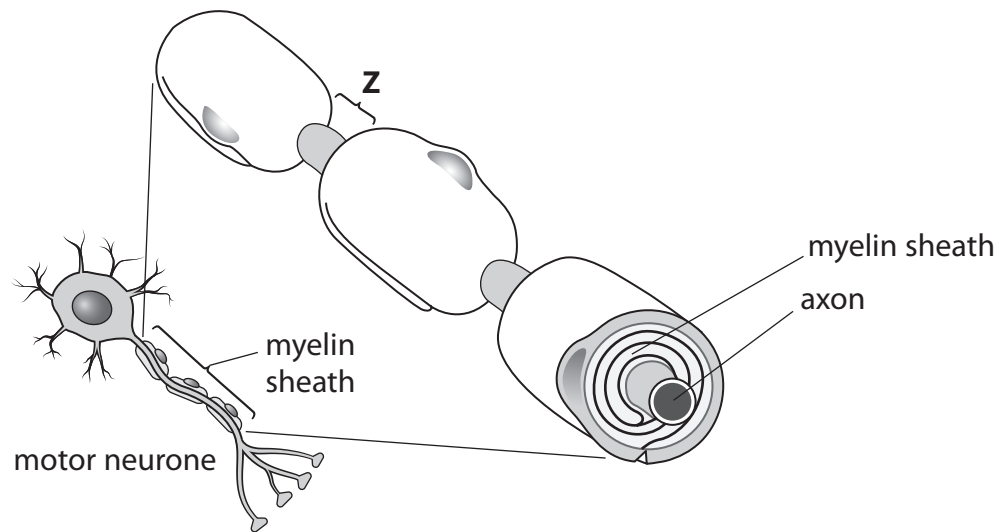
BLANK PAGE



P 5 3 9 7 9 A 0 7 1 2

3 Figure 4 shows a healthy myelinated motor neurone.

A myelin sheath surrounds the axon of the myelinated motor neurone.



© BlueRingMedia/Shutterstock

Figure 4

(a) Name the region labelled Z in Figure 4.

(1)

(b) Describe the structure of the myelin sheath.

(4)



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA



DONOTWRITEINTHISAREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

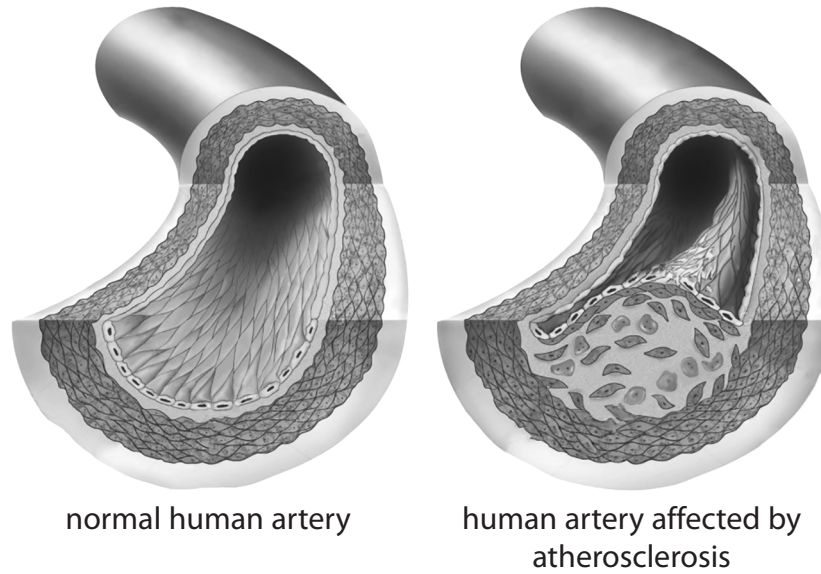
DO NOT WRITE IN THIS AREA



- 4 Smoking is a risk factor for the development of the disease atherosclerosis.

Damage to the artery wall may lead to atherosclerosis.

Figure 6 shows a part of a normal human artery and a part of an artery with atherosclerosis.



© Encyclopaedia Britannica/UiG/Getty Images

Figure 6

Discuss how smoking is a risk factor for the development of atherosclerosis.

(6)

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Handwriting practice area with 20 horizontal dotted lines.

(Total for Question 4 = 6 marks)

TOTAL FOR PAPER = 30 MARKS



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

BLANK PAGE

